Boundary handling in Wiener-AR identification

Controlled numerical checks and implementation decision

9 November 2026

1. Main findings and scope

The earlier BI-HMC and optimization comparisons used different assumptions at the beginning of the retained data record. BI-HMC replayed observed input prehistory and included an initial AR-noise covariance contribution. The optimization methods restarted their fitting filters at zero after the data-generation burn-in. This difference explains a substantial part of the observed parameter-error advantage in the two tested benchmarks.

Two directions of intervention support that interpretation. Giving the FSM criterion input-history replay and a conditional AR likelihood brought its well-optimized estimates close to BI-HMC. Conversely, removing prehistory and initial-covariance treatment from BI-HMC moved its posterior-mean errors close to the zero-state optimization criterion. Priors and the preliminary posterior optimization were retained in the reverse test.

These are finite-record numerical findings. They establish neither general Bayesian superiority nor asymptotic inconsistency of the optimization methods. The current implementation deliberately uses a common zero-state fitting convention for comparability; the simulator's burn-in is unchanged.

The main experiments used the saved `example_1` and `example_CSTR` records: 30 independent records per benchmark, 10,000 retained observations per record, 100 discarded generation samples, and innovation standard deviation 0.1. Inputs and outputs were matched by saved hashes. Ground truth was used for assessment, not as an inference input. The error reported in the tables is

$$E_\theta = 100\,\frac{\|\widehat{\theta} - \theta_{\mathrm{true}}\|_2}{\|\theta_{\mathrm{true}}\|_2}.$$

Table entries are arithmetic means of this percentage error across the stated records. They are not componentwise signed bias, parameter RMSE, or posterior credible intervals. The separately estimated innovation variance is excluded from the coefficient-error vector.

2. What differs at the record boundary?

Write the signal and disturbance as

$$\alpha_t = \frac{B(z^{-1})}{A(z^{-1})}u_t$$

$$\beta_t = \alpha_t + \sum_{j=2}^{n_f} f_j \alpha_t^j$$

$$c_t = \beta_t + e_t$$

$$D(z^{-1})e_t = \nu_t.$$

The input is named `r` in Python. The model is strictly causal in its linear signal block, and the innovations are independent zero-mean Gaussian samples.

Fitting protocol	Signal calculation	AR-noise likelihood
Original optimization convention; current common convention	Start the signal filter at zero at the first retained input.	Set pre-record residuals to zero; score all retained innovations.
Former BI-HMC convention	Replay the observed input prefix, then retain the fitted signal over the measurement interval.	Integrate the unknown noise state through its finite-burn-in initial covariance.
Conditional protocol tested in FSM	Replay the same observed input prefix.	Use the first noise-order output observations for conditioning; score subsequent innovations.

Increasing the simulator's burn-in does not make the signal state zero at the retained-record boundary. With an ongoing input, a stable filter generally carries a nonzero state into that interval. Replaying input history reconstructs that candidate-dependent state. It must be recalculated when the candidate coefficients change; initializing it once at the starting parameter vector is not equivalent.

For a candidate coefficient vector, define residuals and reconstructed innovations:

$$e_t(\theta) = c_t - \beta_t(\theta)$$

$$\hat{\nu}_t(\theta) = e_t(\theta) + \sum_{i=1}^{n_d} d_i e_{t-i}(\theta).$$

Under the common zero-state convention, unavailable pre-record residuals are zero. Every retained sample contributes to

$$Q_0(\theta) = \sum_{t=1}^{N} \hat{\nu}_t(\theta)^2$$

$$\log L_0 = -\frac{N}{2}\log(2\pi\sigma_\nu^2) - \frac{Q_0(\theta)}{2\sigma_\nu^2}.$$

The finite-record AR whitening matrix is unit lower triangular, so its determinant is one. The output disturbance is still colored Gaussian noise; zero initial states do not make the output noise white. It is the whitened innovation sequence that is modeled as independent.

For the former finite-start likelihood, let $m=\min(n_d,N)$, and let Γ_d be the covariance of the first m noise residuals divided by innovation variance. The initial part instead contributes

$$Q_{\text{full}}(\theta) = e_{1:m}^{\mathsf{T}} \Gamma_d^{-1} e_{1:m} + \sum_{t=m+1}^{N} \hat{\nu}_t(\theta)^2$$

$$\log L_{\text{full}} = -\frac{N}{2}\log(2\pi\sigma_\nu^2) - \frac{\log|\Gamma_d|}{2} - \frac{Q_{\text{full}}(\theta)}{2\sigma_\nu^2}.$$

Both the residual mean and this covariance depend on candidate coefficients. The conditional FSM experiment used input-history replay but omitted the first n_d likelihood contributions. Here $n_d=1$, so it scored 9,999 innovations. That tested alternative is distinct from the zero-state protocol now adopted in production, which scores all 10,000 retained samples.

3. Numerical evidence

3.1 Separating optimization accuracy, priors, and initial states

The following controls used all 30 matched records per benchmark. Maximum likelihood omitted both coefficient and innovation-variance priors and profiled the variance. Posterior modes used the same physical priors as BI-HMC. Refinement optimized the stated physical objective, without a sampling-coordinate Jacobian term. A second refinement checked for residual numerical improvement.

Procedure, 30 records per benchmark	Example 1 error (%)	CSTR error (%)
Original saved FSM WS-GNI	1.24945	2.25868
Further optimization of the unchanged FSM criterion	1.24945	2.05071
Maximum likelihood with input replay and finite initial-noise covariance	0.68207	0.79182
Physical posterior mode with those full initial-state conventions	0.68251	0.79155
Saved BI-HMC posterior mean	0.68245	0.78663
Physical posterior mode with both states reset	1.24788	2.05552
Maximum likelihood with both states reset	1.24945	2.05072

A well-optimized likelihood fit already reproduces nearly the saved Bayesian mean accuracy when its initial-state conventions match. On these records, posterior averaging and the selected priors are not required to reproduce that average accuracy. This does not rule out their value for uncertainty estimation, shorter records, weaker excitation, or other noise regimes.

For the likelihood controls, changing both boundary conventions reduced mean error by 0.56738 percentage points in Example 1 and 1.25890 points in CSTR. Paired percentile-bootstrap 95% intervals were [0.15741, 1.09255] and [0.57499, 2.08678], respectively, using 20,000 resamples and seed 901109. These quantify variation across records, not posterior uncertainty.

3.2 Which boundary component matters most?

Physical posterior-mode treatment, 30 records	Example 1 error (%)	CSTR error (%)
Signal prehistory and finite initial-noise covariance	0.68251	0.79155
Signal prehistory, but zero initial-noise state	0.68289	0.79216
Signal reset, but finite initial-noise covariance	1.38812	1.75863
Both states reset	1.24788	2.05552

Once signal prehistory is included, changing only the noise-start covariance has a small mean effect: 0.00038 and 0.00061 percentage points. The large effect is associated with the signal restart. When that signal restart is wrong, the two boundary components interact; their effects must not be added independently.

3.3 Literal FSM implementation test

This test retained the archived WS-GNI starting vectors, tuning and budget: 300 stored states, corresponding to 299 updates. It retained the FSM response length of 128 and frequency grid of 1,024, and used the existing autodiff gradient/Hessian and update rule. Only the signal-history replay and conditional residual criterion were changed in an isolated process.

Procedure, 30 records per benchmark	Example 1 error (%)	CSTR error (%)
Original saved FSM WS-GNI	1.24945	2.25868
Conditional boundary-adjusted WS-GNI, original budget	0.68255	1.07168
Same corrected FSM criterion, additional L-BFGS refinement	0.68255	0.79235
Saved BI-HMC posterior mean	0.68245	0.78663

The unchanged baseline reproduced the saved estimates. The conditional FSM loss matched an exact IIR calculation to numerical precision at the checked points. Finite-difference checks of its Hessian had relative discrepancies of approximately 1.34e-9 and 2.13e-10.

The boundary change alone brought Example 1 to the Bayesian mean-error level. CSTR retained numerical optimization error at the original budget: terminal gradient norms were approximately 0.0026-0.0032. Refining that same corrected criterion reduced them below 2.3e-8 and brought its average coefficient error close to BI-HMC. The refinement used no Bayesian prior or posterior optimization.

3.4 Reverse experiment: remove BI-HMC's boundary treatment

The first five saved records of each benchmark were selected before examining the reverse-test outcomes. Both normal and zero-state BI-HMC were rerun on each record with paired sampler seeds. Priors, starting vectors, coordinate jitter, 30-step L-BFGS initializer, adaptation settings and sampling budgets were held fixed within each pair. Starting vectors were preserved for each method, rather than made identical across different methods.

Example 1 retained four chains and CSTR retained one chain. Each chain used 300 warm-up transitions and 500 retained draws. The experiment therefore comprised 20 complete HMC fits. The normal version replayed input history and used finite-start noise covariance; the other version used neither treatment.

Procedure, five matched records per benchmark	Example 1 error (%)	CSTR error (%)
Fresh BI-HMC with its former boundary treatment	0.58645	0.54489
Fresh BI-HMC with signal and noise states reset	1.35521	1.41459
Original saved FSM WS-GNI on those same records	1.35864	1.70701
Well-optimized zero-state maximum likelihood on those records	1.35865	1.40229

Removing the treatment increased mean HMC error by 0.76876 and 0.86970 percentage points. The five-record paired-bootstrap 95% intervals were [0.15224, 1.52306] and [-0.01949, 2.23209]. The CSTR interval includes zero, so this small MCMC subset alone does not establish a precise or

statistically significant population mean increase at the 95% level. Individual records can improve by chance even when the aggregate error rises.

The zero-state HMC results closely follow the zero-state optimization target. The higher original CSTR WS-GNI error includes additional finite-budget optimization error, consistent with the separate refinement controls.

There were no retained divergences. Seven of ten Example 1 fits passed all configured diagnostics; one normal and two zero-state fits exceeded the 1.01 R-hat threshold, with maxima 1.0295 and 1.0115 respectively. CSTR's single-chain runs cannot provide multi-chain R-hat screening. Their minimum mean-ESS values were approximately 348 with normal treatment and 493 with zero states. These are exploratory point-estimation checks, not a claim of fully certified posterior convergence or interval coverage.

The largest relative differences between HMC means and their corresponding physical posterior modes were 0.019% in Example 1 and 0.066% in CSTR, normalized by the mode-vector norm. Their small size relative to the observed error shifts, together with the 30-record target controls, supports a change of fitted statistical target as the main explanation rather than a large sampling error in the reported means.

4. Why a short transient affects the AR coefficient

At the true coefficients, a signal restart leaves a deterministic discrepancy between the actual signal path and the restarted one. The apparent output residual contains that discrepancy as well as AR noise. AR whitening then also acts on the discrepancy, so fitting the noise coefficients can partly fit the transient's temporal structure.

Post-hoc diagnostics at the true signal parameters gave first-sample RMS restart discrepancies of 1.62 in Example 1 and 1.27 in CSTR, against innovation standard deviation 0.1. Most discrepancy energy was concentrated in the first several samples. These diagnostics used truth only for evaluation.

RMSE of the AR coefficient d1 across 30 posterior-mode fits	Example 1	CSTR
Both states reset	0.02316	0.03163
Input history and finite initial-noise covariance	0.00992	0.00960

Signal-block coefficient RMSEs changed much less. For CSTR, the empirical signed mean error in d1 decreased from approximately +0.01867 to +0.00075. For Example 1, dispersion rather than a large signed mean shift accounted for much of the effect. Thus an increase in mean norm error should not simply be called statistical bias.

For a fixed stable model with finite total startup-error energy, its contribution to an averaged criterion can vanish as the record grows. An asymptotic inconsistency claim would require separate analysis of the target, identifiability, excitation, stability and convergence conditions. These finite-record experiments do not supply such a proof.

5. Preliminary optimization is a separate issue

A separate five-record-per-benchmark test compared single-chain HMC with and without the 30-step initializer, keeping 300 warm-up transitions and 500 retained draws. Example 1 mean errors were 0.58537% with initialization and 0.58450% without. CSTR mean errors were 0.54489% and 49.69271%: two uninitialized chains had errors above 123% and effective sample sizes near 3.

Those failures concern finite-budget sampling robustness. They do not show that a converged Bayesian estimator has a 49.7% statistical error. The initializer can prevent poor exploration even though a sufficiently optimized likelihood fit can attain similar point accuracy. It was therefore retained unchanged in both the reverse-boundary experiment and the production change. Its cost belongs in total wall-clock time; HMC transitions are not comparable one-for-one with optimizer updates.

6. Current implementation and rerunning studies

BI-HMC now uses zero signal and AR-noise states at the first retained sample and scores every retained innovation. The input-history API, finite-start and stationary covariance machinery, and conditional-noise option were removed. Obsolete boundary settings raise an error. Priors, smooth stability coordinates, Jacobian compensation, initialization optimization and physical posterior-mean estimation are unchanged.

Saved input prehistory and its hash are retained as data provenance for Python/MATLAB replay checks, without entering inference. Metadata distinguishes the generation count `data_burn_in` from inference's `noise_burn_in=0` and `input_history_length=0`. The policy is named `zero_state_fit_after_burn_in`. The numerical distinction between exact IIR evaluation in HMC and FSM approximation in the optimizers remains.

From the repository root, these commands run both benchmarks and both parametrizations, with FSM first and PL through MATLAB:

```
python python/run_studies.py --study montecarlo --cpu
python python/run_studies.py --study initialization --cpu
python python/run_studies.py --study noise --cpu
python python/run_studies.py --study record-length --cpu
```

Monte Carlo and initialization export matched Python records/starts for PL. Noise and record-length studies retain their existing backend-specific record generators and pairing rules. Add `--dry-run` to preview, or use `--experiment` and `--parametrization` to select a subset. Existing result files were not relabelled; regenerate them before presenting the new convention.

7. Validation, provenance and interpretation limits

The Python implementation passed 46 tests, including dense-Gaussian likelihood checks, gradient/Hessian checks, stability transforms, posterior summaries, real tiny study runs, shared-record export and command-plan validation. Changing prehistory provenance while holding the retained data fixed produced identical posterior draws, verifying that history no longer affects inference.

The MATLAB R2026a retry on 9 November 2026 passed all four Python-to-PL study checks: Monte Carlo, initialization, noise sensitivity and record length. The first two verified shared-record replay and paired exact-sign tests; the latter two verified reuse of saved BI-HMC estimates in PL summaries. The existing shared-BI-HMC alignment, cohort, timing and hypothesis tests also passed. Separate existing checks for posterior/comparison plots, study outcome and runtime reports, and paired parameter-error tests passed as well. That separate plotting batch completed every assertion but stalled during shutdown; its two isolated batch processes were stopped after checking the success markers, without targeting the interactive MATLAB session. The smoke fixtures used two repetitions, 16-24 observations, one optimizer plus BI-HMC, and deliberately tiny sampling budgets. They validate integration, not posterior convergence or experimental accuracy. All fixture files were removed; existing study results were preserved.

The retry exposed a MATLAB singleton-method summary-shape error in the record-length path. Explicit allocation now preserves the parameter axis in record-length and noise statistics. A new regression check passed for both studies, recomputing the single-method summaries independently

and checking the inverse-record-length fit. This is a summary-storage correction, not a change to an optimizer or to the boundary convention. Earlier startup errors 5201 were avoided by allowing normal MATLAB preference and licensing access.

The historical ablations were isolated from production code. Their temporary scripts and outputs were deleted as requested; the numerical summaries above are transcribed from the checked session results, and this report is not a raw chain archive. During the reverse test, the CSTR comparison file was updated by another process. Its starting version was read from Git and verified against the initial checksum; the newly updated file was preserved. The five-record HMC and 30-record control tables therefore refer to the original matched records, not a mixture of result revisions.

The inference is limited to these benchmark settings and representative FSM WS-GNI comparisons. It does not rank every optimizer at every record length, noise intensity or initial parameter vector. Aligning boundary assumptions removes a demonstrated comparison confound; differences in priors, numerical optimization, approximation accuracy and posterior uncertainty remain legitimate subjects for the regenerated experiments.

Related implementation and usage sources: BI-HMC implementation (bayesian_hmc_implementation.md), study design (identification_studies.md), likelihood code (../python/bayesian_hmc.py), experiment launcher (../python/run_studies.py), and repository README (../README.md).